

● EASY

● ALGEBRA

Algebra

10 questions · ~15 min

04

Q1

ALGEBRA

Tom scored 85, 78, and 98 on his first three exams in history class. Solving which inequality gives the score, G , on Tom's fourth exam that will result in a mean score on all four exams of at least 90 ?

A. $90 - (85 + 78 + 98) \leq 4G$

B. $4G + 85 + 78 + 98 \geq 360$

C. $\frac{(G + 85 + 78 + 98)}{4} \geq 90$

D. $\frac{(85 + 78 + 98)}{4} \geq 90 - 4G$

Line k is defined by $y = \frac{1}{4}x + 1$. Line j is parallel to line k in the xy -plane. What is the slope of j ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Henry receives a \$ 60.00 gift card to pay for movies online. He uses his gift card to buy 3 movies for \$ 7.50 each. If he spends the rest of his gift card balance on renting movies for \$ 1.50 each, how many movies can Henry rent?

- A. 10
- B. 25
- C. 35
- D. 40

$$gx = 11x + 4$$

For the given linear function g , which table shows three values of x and their corresponding values of gx ?

A.

x	gx
-1	7
0	11
1	15

B.

x	gx
-1	-4
0	0
1	4

C.

x	gx
-1	-7
0	4
1	15

D.

x	gx
-1	-11
0	0
1	11

Q5

ALGEBRA

$$y = -3x$$

$$4x + y = 15$$

The solution to the given system of equations is x, y . What is the value of x ?

A. 1

B. 5

C. 15

D. 45

During a portion of a flight, a small airplane's cruising speed varied between 150 miles per hour and 170 miles per hour. Which inequality best represents this situation, where s is the cruising speed, in miles per hour, during this portion of the flight?

- A. $s \leq 20$
- B. $s \leq 150$
- C. $s \leq 170$
- D. $150 \leq s \leq 170$

$$y = 70x + 8$$

Which table gives three values of x and their corresponding values of y for the given equation?

A.

x	y
0	8
2	148
4	288

B.

x	y
0	70
2	78
4	86

C.

x	y
0	70
2	140
4	280

D.

x	y
0	8
2	132
4	272

Q8

ALGEBRA

$$k + 12 = 336$$

What is the solution to the given equation?

- A. 28
- B. 324
- C. 348
- D. 4,032

Q9

GRID-IN ALGEBRA

The function f is defined by the equation $fx = 7x + 2$. What is the value of fx when $x = 4$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$10x = 110$$

$$6x - 63 = y$$

The solution to the given system of equations is x, y . What is the value of y ?

A. 63

B. 11

C. 10

D. 3